

The oracle—constructed separately to represent a constant or balanced function—is composed directly into this superposition. After the oracle acts, additional Hadamard gates are applied to the input qubits to transform phase differences introduced by the oracle into measurable computational basis states.

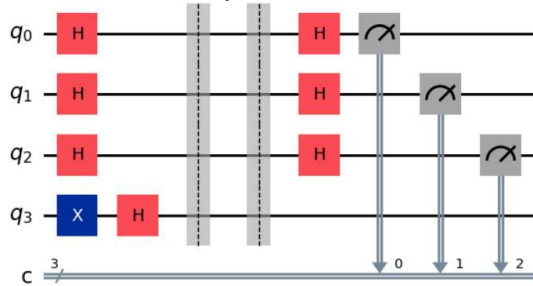


Figure 2: Circuit Diagram for Deutsch-Jozsa Circuit Implementation. Shows initial H gates, oracle block, Final H gates, and measurements

The resulting circuit diagram clearly shows all components of the algorithm: the initial Hadamards, the oracle block, the final Hadamards, and the measurement layer.

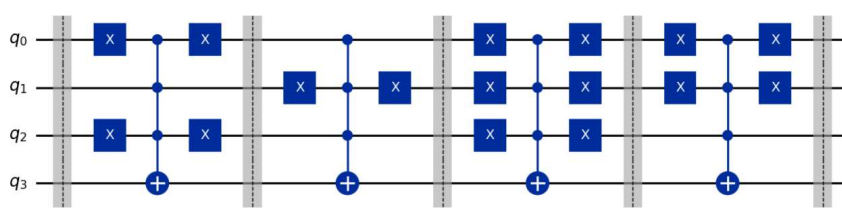


Figure 3: generated by `dj_query(3)` the multi-controlled structure with X gates, barriers, and the target qubit. This is a portion of a circuit before Algorithm was inserted.

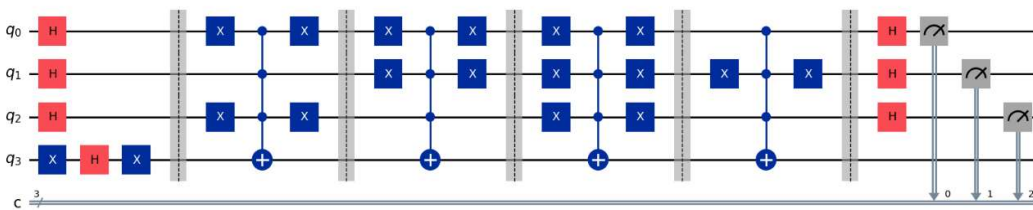


Figure 4: Circuit After compiled with DJ Algorithm We implemented $n=3$ qubits.

Expected Outputs:

If the function is constant, the circuit will always measure 000...0 on the input qubits. If the function is balanced, the output will be anything other than all zeros.

Simulation Result

The circuit was then run on a simulator and the following result was visualised:

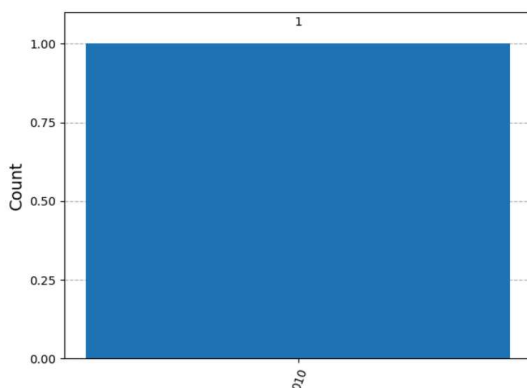


Figure 5: Result simulated on a simulator shown on a histogram. The resulting string is "010"

This shows the expected behavior for a balanced oracle under ideal, noise-free simulation.

Circuit Execution on IBM Quantum

The circuit was then run on real devices such as the IBM Quantum Computer. Firstly, the circuit was transpiled and then run on a hardware such as “ibm_fez” The result it gave us on the hardware is as follows: `{'110': 1}` balanced We then visualized result on a histogram as follows:

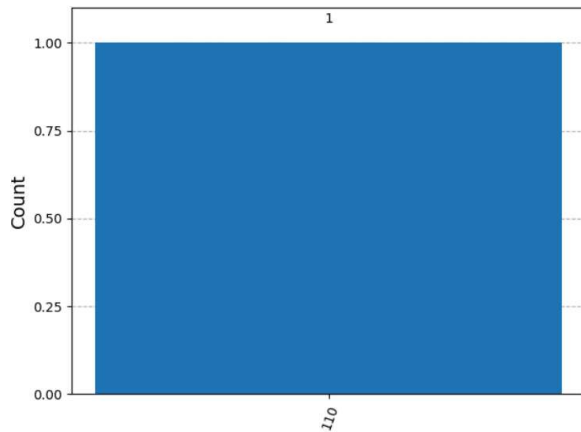


Figure 6: Histogram of result on hardware(ibm_fez)

Comparison: Simulator vs. Hardware

We later compared the result of the simulator and the IBM quantum computer hardware as follows:

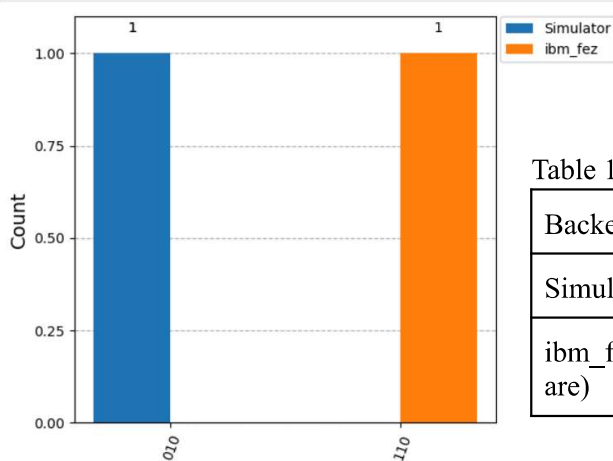


Figure 7: Left histogram is comparison of the results

Table 1: Side by side comparison of the two backends

Backend	Output	Result
Simulator	010	Balanced
ibm_fez(hardware)	110	Balanced

Quantum Noise and Discrepancies

Although the specific measured strings differ, the classification matches perfectly. This may be due to real devices such as the hardware may have introduced noise at many levels including gate errors, decoherence, measurement errors and more.

Conclusion

The experiment demonstrates the key idea behind the Deutsch–Jozsa algorithm: even in the presence of hardware noise, the quantum circuit preserves the essential information needed to distinguish constant and balanced functions.

In our run, the result matched expectations. However, the output differed due to hardware errors. This includes gate errors, decoherence and measurement errors.

The results confirm that the algorithm works as expected. It provides exponential speedup in theory, and quantum noise affects measurement but not the final classification.